

PROFILE

MACHINE LEARNING FOR
RESILIENT, TRUSTWORTHY SYSTEMS

Ph.D. Candidate in Computer Science at Fordham University developing machine-learning methods for DDoS detection and network defense, with connected research in trustworthy AI, computer vision, language, and multimodal intelligence.

EDUCATION

Expected Spring 2027

Ph.D. Candidate in Computer Science
Fordham University · New York, New York
Focus: AI for network security, machine learning, large language models, computer vision, and multimodal systems.

TEACHING EXPERIENCE

Spring 2026

Instructor · CISC 1100 E01, Structures of Computer Science
Fordham University. Discrete structures, logic, relations, functions, combinatorics, graph theory, and computer-based lab work.

Fall 2025

Instructor · CISC 1100 R03, Structures of Computer Science
Fordham University. Set theory, logic, Boolean algebra, recursion, graphs, and rigorous problem solving.

RESEARCH EXPERTISE

AI for network security

DDoS detection and mitigation

Trustworthy and data-efficient AI

Large language models

Computer vision

Multimodal learning

Federated learning

Explainable AI

INTELLECTUAL PROPERTY

Granted · active

Cancer subtype identification via multi-omics data integration
Chinese patent CN113537358B. Co-inventors: Hongmin Cai and Ali Alfatemi.
Patent record ↗

PROFESSIONAL SERVICE
SELECTED PEER-REVIEW SERVICE

Information Fusion · IEEE Transactions on Network and Service Management · Scientific Reports · Artificial Intelligence Review · Signal, Image and Video Processing · The Journal of Supercomputing · Cluster Computing · IEEE WCCI 2024 · IJCNN 2025

PUBLICATIONS

Complete publication record

21 published works · 5 preprints

- 2026 A Resource-Efficient Blockchain With Delegated Fault-Tolerance for Manufacturing Nodes**
Mohammad Iqbal Saryuddin Assaqty, Ying Gao,
Ali Alfatemi
, Mohamed Rahouti, Abdellah Chehri. *IEEE Transactions on Industrial Informatics* · Volume 22, Issue 5, pp. 4340–4349.
[Publication record](#)↗
-
- 2026 MS-GBANet: Multiscale graph convolution with boundary attention for medical image segmentation**
Aiman Solyman, Ahmed Elazab, Mohamed Rahouti,
Ali Alfatemi
, Zeinab Mahmoud, Marco Zappatore. *Biomedical Signal Processing and Control* · Volume 121, Article 110397; issue date 1 August 2026.
[Publication record](#)↗
-
- 2026 MTAGEC: Multi-Task Arabic Grammatical Error Correction as a sequence generation with synthetic data**
Zeinab Mahmoud, Marco Zappatore, Natalia Kryvinska, Mohammed Abdalsalam,
Ali Alfatemi
, Aiman Solyman. *Journal of King Saud University Computer and Information Sciences* · Volume 38, Issue 2, Article 35.
[Publication record](#)↗
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- 2026 ShallowNet: A Lightweight Neural Network Approach for Efficient Flow-Level DDoS Detection**
Ali Alfatemi
, Mohamed Rahouti, Abdellah Chehri, Zakirul Alam Bhuiyan. *IEEE Transactions on Network and Service Management* · Volume 23, pp. 5940–5949; published 10 July 2026.
[Publication record](#)↗
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- 2026 Vision-Language Integration for Image Captioning Using Vision Transformers and GPT-J**
Ali Alfatemi
, Mohamed Rahouti, Mohammed Aledhari, Nasir Ghani, Abdellah Chehri, Gwanggil Jeon. *Applied Imagery Pattern Recognition (AIPR 2025)* · AIPR 2025 proceedings, LNCS 16446, pp. 101–114; published in 2026.
[Publication record](#)↗
-
- 2025 A Two-Stage LLM-Enhanced DDoS Detection Framework for Next-Generation IoT and Edge Networks**
Ali Alfatemi
, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, Aiman Solyman. *IEEE GLOBECOM 2025* · CISS Symposium, pp. 7–12.
[Publication record](#)↗
-
- 2025 Enabling Real-Time, Explainable DDoS Mitigation via On-Premise Large Language Models and Flow Analysis**
Henok Wondimu,
Ali Alfatemi
, Mohamed Rahouti, Abdellah Chehri, Paweł Weichbroth, Nasir Ghani. *Procedia Computer Science* · Procedia Computer Science 270, pp. 1390–1399.
[Publication record](#)↗
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- 2025 Enhancing DDoS Detection for Edge Networks With Noise-Tolerant Shallow Neural Networks in Computational Social Systems**
Ali Alfatemi
, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, Mohammed Aledhari, Gwanggil Jeon. *IEEE Transactions on Computational Social Systems* · Early Access, pp. 1–14.
[Publication record](#)↗
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- 2025 Foreground-Centric learning improves robustness in fine-grained visual recognition**
Ali Alfatemi
, Mohamed Rahouti, Majjed Al-Qatf, Senthil Kumar Jagatheesaperumal. *Signal, Image and Video Processing* · Volume 19, Issue 14, Article 1178.
[Publication record](#)
-
- 2025 Identifying Distributed Denial of Service Attacks Through Multi-Model Deep Learning Fusion and Combinatorial Analysis**
Ali Alfatemi
, Mohamed Rahouti, D. Frank Hsu, Christina Schweikert, Nasir Ghani, Aiman Solyman, Mohammad I. Saryuddin Assaqty. *Journal of Network and Systems Management* · Volume 33, Issue 1, Article 8.
[Publication record](#)
-
- 2025 Integrating Robotic Navigation with Blockchain: A Novel PoS-Based Approach for Heterogeneous Robotic Teams**
Nasim Paykari,
Ali Alfatemi
, Damian M. Lyons, Mohamed Rahouti. *arXiv* · arXiv:2505.15954, posted in 2025; presented at UR 2024.
[Publication record](#)
-
- 2025 ProtoMAML: A Hybrid Meta-Learning Approach Integrating Prototypical Networks for Data-Efficient DDoS Attack Detection**
Ali Alfatemi
, Mohamed Rahouti, Zakirul Alam Bhuiyan, Aiman Solyman, Mohammed Aledhari. *IWCMC* · IWCMC 2025, pp. 1144–1149.
[Publication record](#)
-
- 2024 Advancing DDoS Attack Detection: A Synergistic Approach Using Deep Residual Neural Networks and Synthetic Oversampling**
Ali Alfatemi
, Mohamed Rahouti, Ruhul Amin, Sarah ALJamal, Kaiqi Xiong, Yufeng Xin. *arXiv* · arXiv:2401.03116.
[Publication record](#)
-
- 2024 Advancing NCAA March Madness Forecasts Through Deep Learning and Combinatorial Fusion Analysis**
Ali Alfatemi
, Mohamed Rahouti, D. Frank Hsu, Christina Schweikert. *Intelligent Systems Conference* · IntelliSys 2024, LNNS 1067, pp. 539–560.
[Publication record](#)
-
- 2024 Enabling Trustworthy Federated Learning in Industrial IoT: Bridging the Gap Between Interpretability and Robustness**
Senthil Kumar Jagatheesaperumal, Mohamed Rahouti,
Ali Alfatemi
, Nasir Ghani, Vu Khanh Quy, Abdellah Chehri. *IEEE Internet of Things Magazine* · Volume 7, Issue 5, pp. 38–44.
[Publication record](#)
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- 2024 Multi-Label Classification with Deep Learning and Manual Data Collection for Identifying Similar Bird Species**
Ali Alfatemi
, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, Abdellah Chehri. *Procedia Computer Science* · Procedia Computer Science 246, pp. 558–565.
[Publication record](#)
-
- 2024 Optimizing Control Strategies for Wheeled Mobile Robots Using Fuzzy Type I and II Controllers and Parallel Distributed Compensation**
Nasim Paykari, Razieh Jokar,
Ali Alfatemi
, Damian Lyons, Mohamed Rahouti. *arXiv* · arXiv:2409.17161.
[Publication record](#)
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- 2024 Precision DDoS Detection Through Gaussian Noise-Augmented Neural Networks**
Ali Alfatemi
, Diogo Oliveira, Mohamed Rahouti, Abdelatif Hafid, Nasir Ghani. *NoF* · NoF 2024, pp. 178–185.
[Publication record](#)
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- 2024 Redefining DDoS Attack Detection Using a Dual-Space Prototypical Network-Based Approach**
Fernando Martinez, Mariyam Mapkar,
Ali Alfatemi
, Mohamed Rahouti, Yufeng Xin, Kaiqi Xiong, Nasir Ghani. *ICCCN* · ICCCN 2024, pp. 1–9.
[Publication record](#)
-
- 2024 Refining Bird Species Identification Through GAN-Enhanced Data Augmentation and Deep Learning Models**
Ali Alfatemi
, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, Ruhul Amin, Abdellah Chehri. *Procedia Computer Science* · Procedia Computer Science 246, pp. 548–557.
[Publication record](#)
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- 2024 Toward Utilizing Bidirectional Multi-head Attention Technique for Automatic Correction of Grammatical Errors**
Zeinab Mahmoud, Natalia Kryvinska, Mohammed Abdalsalm, Aiman Solyman,
Ali Alfatemi
, Ahmad Musyafa. *IAENG IJCS* · IAENG International Journal of Computer Science 51(11), pp. 1886–1897.
[Publication record](#)
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- 2024 Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data**
Mohammed Aledhari, Mohamed Rahouti,
Ali Alfatemi
. *arXiv* · arXiv:2411.05880.
[Publication record](#)
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- 2023 Optimizing the Impact of Data Augmentation for Low-Resource Grammatical Error Correction**
Aiman Solyman, Marco Zappatore, Wang Zhenyu, Zeinab Mahmoud,
Ali Alfatemi
, Ashraf Osman Ibrahim, Lubna Abdelkareim Gabralla. *Journal of King Saud University – Computer and Information Sciences* · Volume 35, Issue 6, Article 101572.
[Publication record](#)
-
- 2023 PBEMS: A Permissioned Blockchain-based Equipment Maintenance System in Smart Manufacturing**
Mohammad Iqbal Saryuddin Assaqty, Ying Gao,
Ali Alfatemi
, Ashraf Osman Ibrahim, Anas W. Abulfaraj, Faisal Binzagr, Fezan Nabawi, Mohammad Reza Fahlevi. *Preprints* · Preprints.org version 2, posted 28 October 2023; not peer reviewed.
[Publication record](#)
-
- 2023 Semi-Supervised Learning and Bidirectional Decoding for Effective Grammar Correction in Low-Resource Scenarios**
Zeinab Mahmoud, Chunlin Li, Marco Zappatore, Aiman Solyman,
Ali Alfatemi
, Ashraf Osman Ibrahim, Abdelzahir Abdelmaboud. *PeerJ Computer Science* · PeerJ Computer Science 9, e1639.
[Publication record](#)
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- 2022 Patient subgrouping with distinct survival rates via integration of multiomics data on a Grassmann manifold**
Ali Alfatemi
, Hong Peng, Wentao Rong, Bin Zhang, Hongmin Cai. *BMC Medical Informatics and Decision Making* · 22(1), 190.
[Publication record](#)
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PROFILES

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